2026 Shenzhen International FinTech Competition

International Track | Topic F Agent Wallet

mandate.

Execution Evidence

Recorded payments, transaction register and verification guide

ChainSentinel Bangladesh

TEAM MEMBERS

MD ARIF SHEKH — Team Leader
Computer Science and Engineering | Undergraduate Year 4
International University of Business Agriculture and Technology (IUBAT)

MD MEHERAB HOSSAIN TALUKDER — Team Member
Computer Science and Engineering | Undergraduate Year 4
International University of Business Agriculture and Technology (IUBAT)

IRIN KHATUN — Team Member
Accounting & Information Systems | Undergraduate Year 4
Jagannath University

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Local chain and Ethereum Sepolia | Synthetic bills and MDT test tokens

01 Evidence Summary

This report is the measurement record for Mandate: the approved task, the saved public transactions, the tested outcomes and their limits. It provides the hashes and evidence locations needed to verify each claim. The companion Technical Report explains the implementation; the Security Self-Assessment assesses its safeguards.

Selected topic and scenario

The recorded scenario is Orbit Studio supplier payments under International Track Topic F. Public evidence uses Ethereum Sepolia, chain 11155111; local evaluation uses disposable chain 31337. MDT has six decimal places and no cash value. Organizer acceptance of Sepolia evidence is unresolved.

Recorded item	Outcome
Approved bill total	425 MDT across HOST-1001, ART-2001 and HOST-1002
Completed batch	350 MDT: HOST-1001 200; ART-2001 150
Deferred bill	HOST-1002 75 MDT; cumulative 425 would exceed the 400 MDT UTC-day cap
Vault balance and reserve	1,000 → 650 MDT; 300 MDT minimum reserve preserved
Day-of-demo spending	350 MDT spent; 50 MDT daily allowance remained at that checkpoint
Final authority	Agent/address window cleared; policy version 7; vault unpaused
Public register	13 original hashes: 11 owner transactions and two agent payments; one owner deposit reverted

Deployment and identity

MDT token 0x61E52ad4912FA0f829d0f1B0571aBc3bc157Ba88

Mandate vault 0x1E2bC947b455DE5825caDE9D44e7Ec04B8B76f89

Owner: 0xDc2341ae41c44400128965aB0770567922F8d6ee
Recorded payment agent: 0xE9987a361f8286C396B0519aeC3AB5D29F4553A4

This is saved evidence, not a new live payment run. Amounts and daily allowance describe the 19 October checkpoint; the daily cap resets at UTC midnight. User MetaMask operations are documented in the setup records. The CLI export's realWalletExtensionTested=false flag is not an automated extension-suite result. [E1, E2]

From Request to Matching Receipts

Source review signed approval and Resume

Three drafts were reviewed against their source and exact recipient registry. The owner signed approveBills for 200,000,000; 150,000,000; and 75,000,000 MDT base units at block 11948984. A separate Resume at block 11949011 activated payments under policy 6. Approvals set permission; they did not transfer MDT. The rules were 250 MDT per bill, 400 per UTC day and a 300 MDT reserve. [E1, E2]

Three live Groq requests

Audit time UTC on 19 October	Request and observed result
15:31:34.791	Approved-total question with explicit no-payment instruction: no transfer; app showed 350 MDT eligible instead of the requested 425 MDT approved. Answer-quality miss retained.
15:35:03.290	Pay that one: focused clarification requested an invoice/bill identifier; no selected bill or payment.
15:37:21.549	Pay all approved bills within my rules: exact plan selected two bills for 350 MDT and deferred 75 MDT. The user then separately confirmed execution.

The times above are observation/verification timestamps. The completed plan is 2ff02e4e-521d-4cb0-acd6-1cceea3ae902; its saved entries link the payment request to both transaction records. [E1, E2]

Exact payments at policy 6

HOST-1001 200 MDT = 200,000,000 base units
Recipient: 0x1BA06f96305366D517E21F5D5aD871Ba154FFA0a
Bill key: 0xfe6186d372a90b599be992390a4f2d5f264592197739724f01a20e9df3f26029
Block 11949115; successful BillPaid and token Transfer matched.

ART-2001 150 MDT = 150,000,000 base units
Recipient: 0xFDC4D1bb2319D04848faFaee5c8CF5A0172E0203
Bill key: 0xc16064a320634dd3cb0dee56e35a5b67133f719f021cf6e2e24d4a19de61da14
Block 11949117; successful BillPaid and token Transfer matched.

The 15:41:59.103 UTC payment audit compared the vault and both recipient balances with block 11949114. Vault decrease was exactly 350,000,000 units; recipients increased by 200,000,000 and 150,000,000. HOST-1002 remained unpaid. Full payment receipts and decoded events are preserved in E2; the register provides both explorer links.

02 Transaction Register Part One

All 13 transactions use Sepolia chain 11155111. Times below are saved preparation times on 19 October 2026 in UTC, not transaction inclusion times. Blocks/status come from the original receipts and final export audit. Full block hashes and gas fields are in transaction-register.json/CSV. [E1, E3]

01 Deploy 1,000 MDT test tokens

Prepared 11:20:08.814 UTC | Owner | Block 11948582 | SUCCESS

0xd79c6a0fc89ec8440e71350f84145f51addb3b00a8d923070b23c06592b73d39

02 Deploy the owner-controlled vault

Prepared 13:53:25.428 UTC | Owner | Block 11948602 | SUCCESS

0x2b53c38d45a186ef647ead9ac6f361839ad9f860150e6fa451370de20b2081e3

03 First vault deposit attempt

Prepared 13:56:26.135 UTC | Owner | Block 11948616 | REVERTED; no MDT transfer

0x715e2f6bf7da5e87946c0f27bdc5bc786339259c7686d438b74fe2017f657bf1

04 Fund the vault with 1,000 MDT

Prepared 14:01:38.150 UTC | Owner | Block 11948673 | SUCCESS

0xac511b822fc144485938651b88365b0216ae5deb03241c5de318f7c32078ee14

05 Set Hostline’s test recipient

Prepared 14:11:08.108 UTC | Owner | Block 11948695 | SUCCESS

0x5ee96ac9a18b63d8f605947746c2bb5180f9273c6af433df73381cd533f5fb60

06 Set Artline’s test recipient

Prepared 14:21:48.240 UTC | Owner | Block 11948737 | SUCCESS

0x1a46a18cfe200fdc8416ce38ae5ee461159a4317da183afe62a9dd9f30145110

The cancelled unsent review and the accounting-only 54,813 wei adjustment have no new transaction hash and are excluded from this mined-transaction register. The original records remain preserved. [E1, E2]

Transaction Register Part Two

Register continued in block order. Timestamp meanings and status conventions are defined on the preceding register page. Full receipt references are in the ZIP index. [E1, E3]

07 Grant the separate agent 24-hour access

Prepared 14:23:33.920 UTC | Owner | Block 11948749 | SUCCESS

0x2c53a2312596cfe2a378b7d53f18291be343fa3839712a6651eaf7c6ea45d82f

08 Fund the agent with 0.00005 Sepolia ETH

Prepared 14:43:40.888 UTC | Owner | Block 11948852 | SUCCESS

0xf4d18e790a061f97025fd07e7d087118d9fa2d70f751d653d885d60a927e85e3

09 Approve 3 reviewed bills

Prepared 15:12:03.321 UTC | Owner | Block 11948984 | SUCCESS

0x9cdbddee454d4b5b48de87ccec7177490406fd471e363d6738b5a35b8b97f5a3

10 Resume agent payments

Prepared 15:17:10.943 UTC | Owner | Block 11949011 | SUCCESS

0xfc7bd570cc2bcb407283eff73c99444124491126a92975021e966d23d724a63c

11 Pay HOST-1001

Prepared 15:38:51.742 UTC | Agent | Block 11949115 | SUCCESS

0xa35b0bd2785c1b9272e571d2a5197eb5530089151c6b8e751935b02941931ba4

12 Pay ART-2001

Prepared 15:39:22.124 UTC | Agent | Block 11949117 | SUCCESS

0x37ba350085b7cf15423745251aa71c8530c7c5915493534663f7eb0a46029ac9

13 Revoke agent access

Prepared 15:47:10.100 UTC | Owner | Block 11949169 | SUCCESS

0xa6a4665cdfe4e0e44e8e08103ef8e943d8ac05fcbe432fef225e940c1dcb60e0

Owner Stop Rejections and Recorded Costs

Confirmed owner revocation

The revokeAgent transaction at block 11949169 emitted AgentRevoked and PolicyChanged(7). The 15:51:50.619 UTC audit found zero agent/start/expiry, policy 7, 650 MDT and unchanged paid flags. The vault was unpaused. The original revocation receipt and detailed event/state checks are in E2. [E2, E3]

Unmined eth_call	Checked result and scope
Duplicate HOST-1001	BillAlreadyPaid at policy 6; no transaction sent and zero gas spent.
Deferred HOST-1002	DailyLimitExceeded at policy 6; unpaid 75 MDT exceeded the remaining 50 MDT. No transaction sent.
Former agent after revocation	OnlyAgent at policy 7; no transaction sent. The same 75 MDT bill also exceeded daily allowance, so this does not isolate an otherwise-eligible public revocation case.

The decoded OnlyAgent error establishes the actual authority rejection in that call. A clean otherwise-eligible revocation scenario is separate local evidence. These three calls are not mined failures and are not added to the 13 transaction count. Two confirmations are the public acceptance threshold, not an absolute finality guarantee. [E2, E4]

Wallet recovery observations

The first protected deposit reverted. Two later protected owner transactions required specific audits of their original hash, inner call, events, code/state and cost. Automatic exact matching was not relaxed. The protected agent grant exceeded its direct-call reservation by 54,813 wei; a local accounting-only record covered the already-spent excess under the aggregate cap. No resend was created by that adjustment. [E2]

Final fee and outflow accounting

Recorded quantity	ETH
Owner gas including reverted deposit	0.00000000032971641
Agent payment gas	0.000000000011372928
Combined owner and agent gas	0.000000000341089338
Owner outflow including 0.00005 ETH agent funding	0.00005000032971641
Conservative maximum reservations	0.000050000761755467
User-approved aggregate cap	0.001

Agent gas is paid from the original allocation and counted once in owner outflow. Its recorded unused balance was 0.000049999988627072 ETH with vault authority revoked. ETH gas and funding are separate from MDT spending/reserve. The values above are historical observations. [E2]

Local Evaluation and Hosting Evidence

Fixed cases model and denominators

The 19 October Groq v5 run measured 40 fixed synthetic cases: 20 ordinary and 20 hostile/exception cases. Six interaction cases are tags within those 40. Each case resets a disposable chain 31337; expected labels are not sent to the model. The run used openai/gpt-oss-120b, prompt mandate-step5-v3, temperature 0.2, low reasoning, 2,000 output tokens, 20-second timeout and at least 15-second call spacing. There was no automatic retry or fallback. [E4]

Measure	Observed
Ordinary completions	19/20
Exception explanations	20/20
Critical extraction fields	156/156 across 26 explicit synthetic CSV invoices
Whole ordinary invoices exact	26/26
Defined interactions	6/6
Unauthorized/mismatched payment cases	0/40
Model calls and provider errors	72 calls; 0 errors
Verified local payment receipts	27 across independent case resets

Earlier failures informed repairs; v5 is a fixed-case regression, not untouched held-out accuracy. N13 recognized an already-paid bill but asked an unnecessary question and failed completion. It remains in the denominator; no duplicate transfer occurred. These results do not establish general PDF/OCR accuracy or unaided-model performance. Raw v5 run, analysis, cases and frozen settings are included in the ZIP. [E4]

Comparison and software-check limits

The automated manual baseline used 24 per-bill signatures; the agent workflow used 19 batch-approval signatures and 20 execution confirmations plus conversation. Timings excluded human reading/signing; no human time saving is established. A virtual-only 20-case hostile comparison had zero unauthorized outflow on both sides, so it did not measure incremental contract benefit. Independent direct-bypass tests cover contract enforcement. [E4]

The isolated 19 October review passed 58 contract, 4 identity, 35 app, 59 agent, 9 evaluator, 11 local Sepolia-preparation and 19 browser checks. These overlap earlier suites and do not expand the 40-case denominator. Source tests can overwrite curated artifacts; reproduce them in an isolated copy. [E4]

Hosted inspection checkpoint

On 20 October, 16 hosting checks and matching Linux CI passed. Render browser sign-in/out and an actual restart retained three reviewed bills and two original confirmed receipts in encrypted Neon storage. The recorded vault balance was 650 MDT and policy 7 remained revoked. This is persistence/access evidence, not a new payment or model-evaluation run. [E5]

03 Reproduction Guide

Verify the archive and transaction evidence

Extract Mandate-Transaction-Logs.zip. Start with README.md, transaction-register.csv and receipt-index.json. SHA256SUMS.txt and manifest.json identify every archive entry. Original saved evidence bytes retain their repository paths. Historical embedded paths resolve through evidence/locations.json; code referenced by freezes belongs to the separate source deliverable.

Open each register explorer link and compare Sepolia network, original hash, receipt status, block number/hash and sender/target. For supplier payments, verify the exact MDT Transfer and BillPaid key, recipient, amount and policy 6. For the owner stop, verify AgentRevoked and PolicyChanged(7). A deployment receipt or approval alone is not supplier-payment proof.

Eleven transactions have detailed saved receipt payloads with logs. Token/vault deployment receipts are preserved as summaries in the original export; a 20 October read-only RPC lookup returned HTTP 403, so no new detailed copy is claimed. receipt-index.json distinguishes both formats and points to the original file/JSON location. Explorer links support further independent checks.

Inspect the hosted records

Open Mandate

Follow competition/JUDGE_REVIEW.md for the inspection route and private access. The record is historical; a new UTC day changes the displayed daily allowance. [E5]

Reproduce locally

Use the Technical Report’s Implementation and Setup section, then the source README for complete check commands. Run in a fresh extraction so generated tests cannot overwrite curated evidence. The manual demonstration is labeled scripted; live-model performance is established by the saved evaluation, not a fresh local replay. [E4]

Evidence sources and retained limits

[E1] evidence/sepolia/public/step5-sepolia-2026-10-19T15-52-14-777Z.json and step5-sepolia-export-verification.json. Audit: 19 October 15:53:34.155 UTC. Export SHA-256: fb8ee64852392ce7c99255ba1520427ed7fc0899c692baec8779c8ed358331b7

[E2] Original public JSON files in evidence/sepolia/public/: setup/recovery, owner review, bill approval, Resume, read-only/clarification/plan, payments and revocation. Full filenames and hashes are in the ZIP manifest.

[E3] transaction-register.json/CSV and receipt-index.json in the ZIP are derived navigation records, not altered originals. [E4] demo/cases.json, demo/evaluation-freeze-groq-v5.json, v5 raw run/analysis and evidence/maintenance/2026-10-19/full-review/. [E5] evidence/hosting/2026-10-20/live/restart-verification.json and reviewer-access/access-verification.json; hosting CI logs. All are indexed in the ZIP.

Original records can contain earlier pending flags or CLI extension-test defaults. Later dated verifications establish the final outcome. Sepolia acceptance and final submission remain pending.